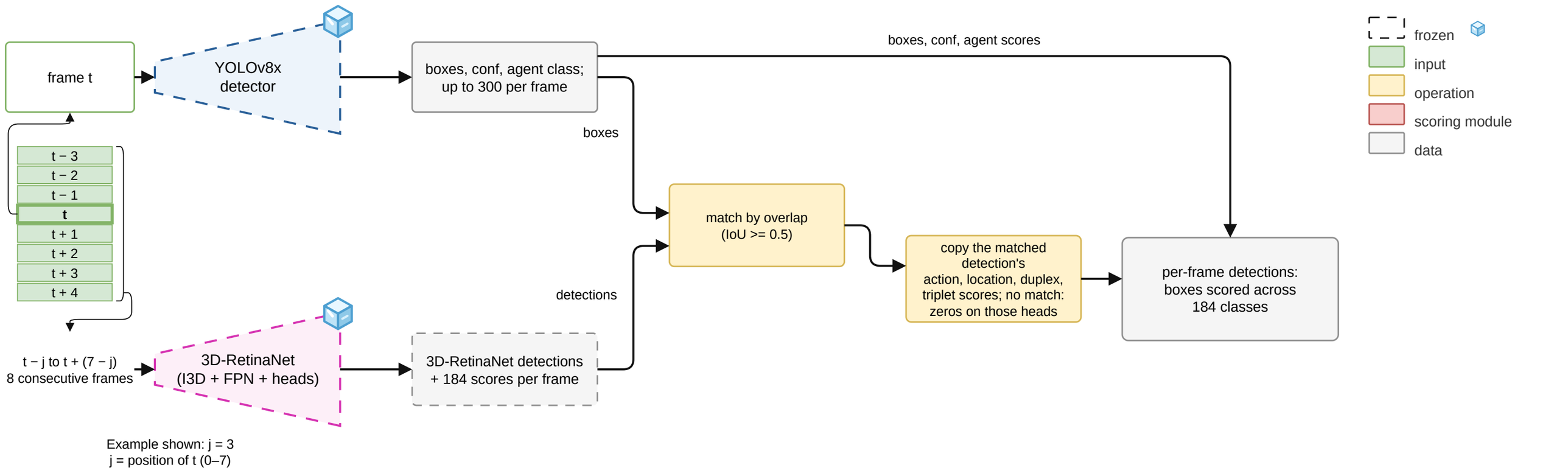

frame t
YOLOv8x
detector
boxes, conf, agent class;
up to 300 per frame
boxes, conf, agent scores
boxes
t − 3
t − 2
t − 1
t
t + 1
t + 2
t + 3
t + 4
match by overlap
(IoU >= 0.5)
copy the matched
detection's
action, location, duplex,
triplet scores; no match:
zeros on those heads
per-frame detections:
boxes scored across
184 classes
detections
t − j to t + (7 − j)
8 consecutive frames
3D-RetinaNet
(I3D + FPN + heads)
3D-RetinaNet detections
+ 184 scores per frame
Example shown: j = 3
j = position of t (0–7)
frozen
input
operation
scoring module
data